

VNUHCM - University of Science
Faculty of Information Technology

CS486 – Lecture 1

Overview of Database Systems

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Content

- **Introduction**
- The evolution of database systems
- Characteristics of the database approach
- Database users
- Architecture of a DBMS
- DBMS category
- Data models
- Database languages

Application of DB systems

- Example of
 - Youtube database
 - ATM databasse
 - Car database
- Unpack them to understand these operators
 - Read
 - Modify
 - Learn



Home



Explore



Shorts



Subscriptions



Library

FILTERS

#1 Vietnamese Flower Shop - Flower Delivery in Vietnam

Flowers & Bouquets Delivery. #1 Flower Delivery in Vietnam. Bouquets from 199k.

Romantic Flowers

Giant Bouquets

Send Flower To Vietnam

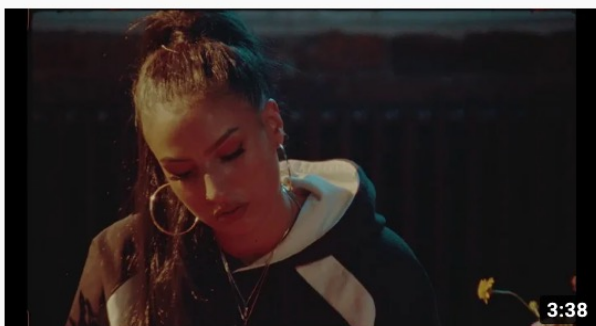
Happy Anniversary

Ad <https://www.flowerstore.vn/>

Upload video

Go live

VISIT SITE



Phoenix Laoutaris - Flowers (Official Video)

261K views • 1 year ago



Phoenix Laoutaris

Stream Flowers" here: https://open.spotify.com/track/4hbNgW2qAaiAbtHe9OILv2?si=OpDCfYFMSzWVpSsgr_xV8Q Come follow ...



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Taylor Swift
3.1M views • 5 days ago
New



How to make Water Hyacinth from paper| Hướng dẫn làm ho...

Kay's Tutorials Channel
1.6K views • 1 year ago



Bố Tinna Tình hát Cát Bụi Cuộc

Đời Sống Tâm Linh
15K views • 5 days ago
New

Youtube database

■ Read

- List of videos and their descriptions
- # views and likes/dislikes

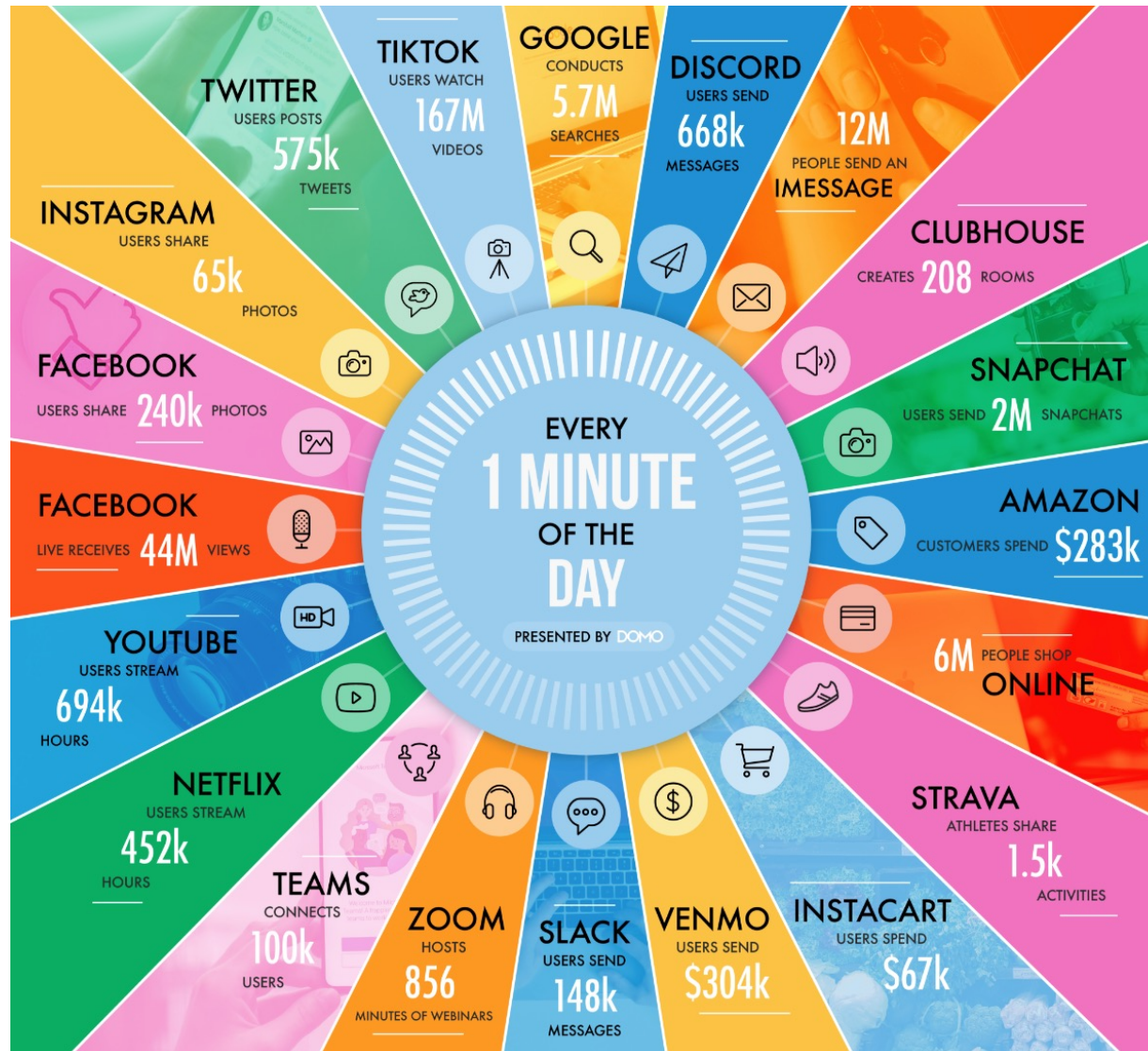
■ Modify

- Videos uploading
- Comments adding
- Likes/dislikes counting

■ Learn

- Related videos
- Relevant ads

An Internet minute in 2021



Source: visualcapitalist.com

Self-driving cars



Source: tesla.com

Car database

■ Read

- Front panel metrics
- Speed, distance
- Estimated time of adrrival

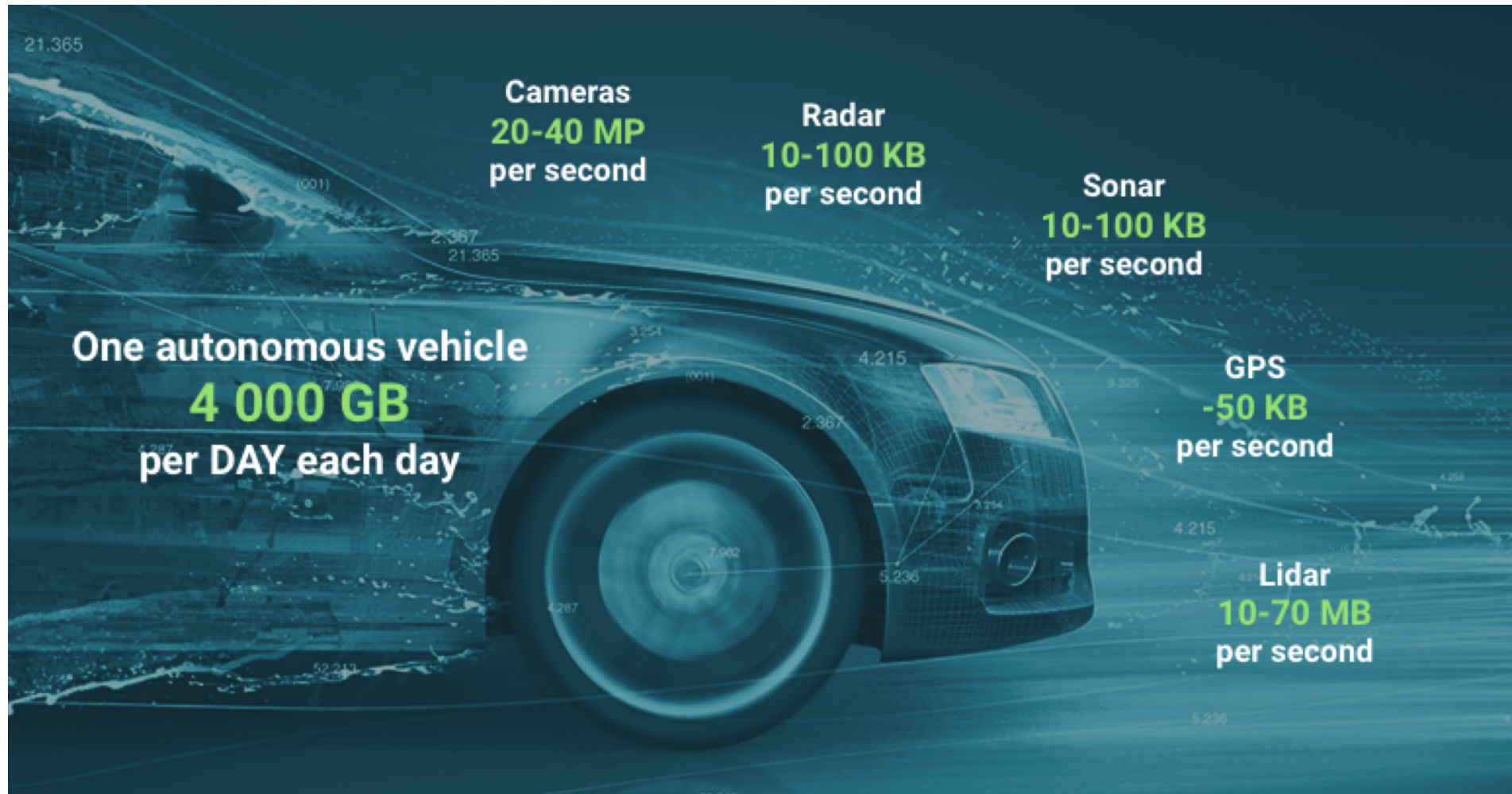
■ Modify

- Road model
- Drive model

■ Learn

- Collisions
- Traffic signals

Flood of data in autonomous vehicles



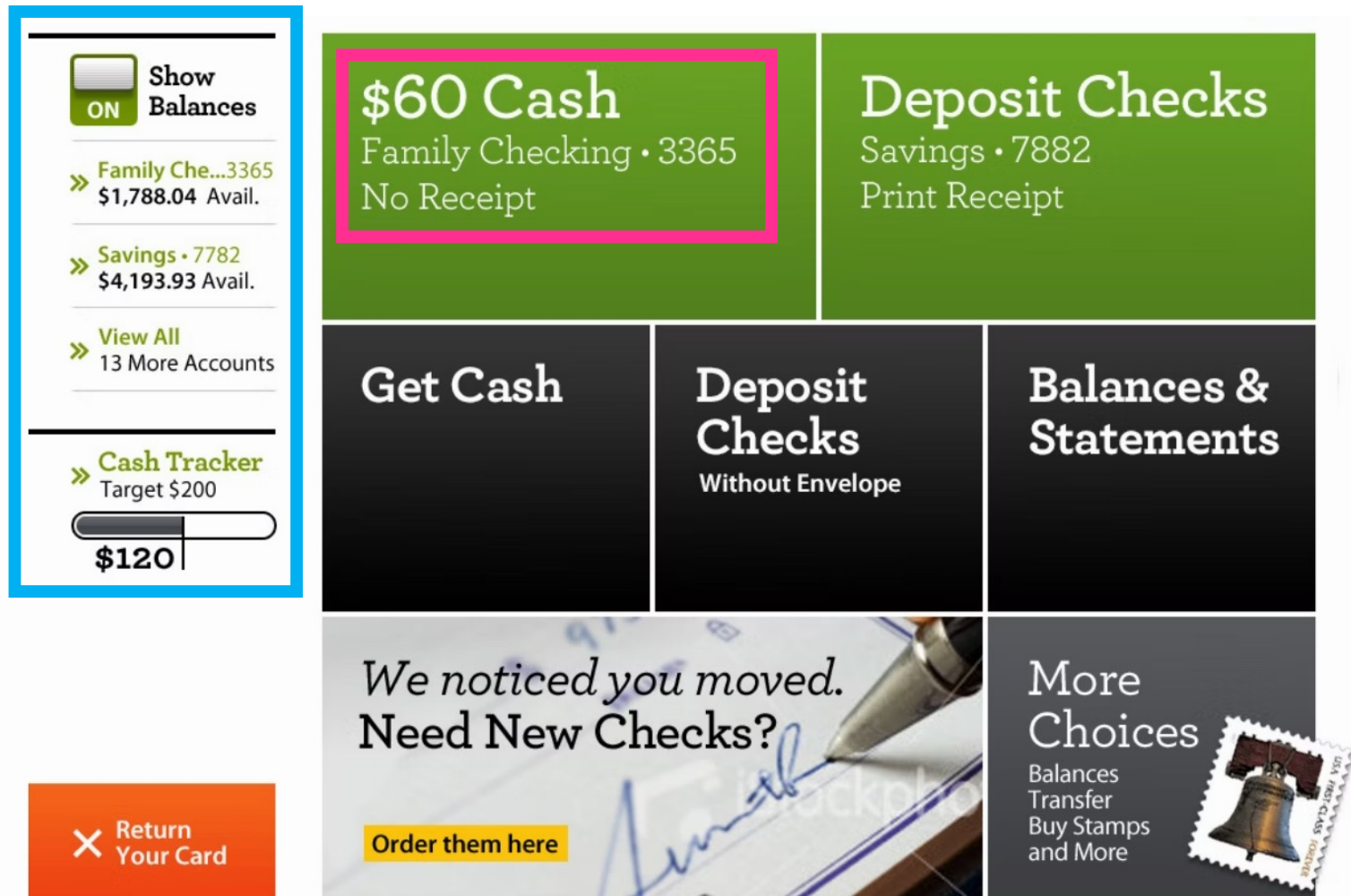
Source: intellias.com

How do bank ATMs work?



Source: wells Fargo.com

Wells Fargo's ATM interface



Source: wells Fargo.com

Example

- We withdraw \$60
- What steps the ATM goes through?
 - Read balance
 - Give money
 - Update balance
- vs
- Read balance
 - Update balance
 - Give money
- How would we design a system around this?

Database (DB)

- A platform to store and manage data
 - To read data, modify data, learn on data
- Good platforms also support
 - Scale, speed, stability
 - Evolution
 - Reliability
 - Cost efficiency

Database (DB)

- Is a collection of related data
- Contains information relevant to an enterprise
- Example: Corporate records
 - Sale/Purchase
 - Payable and receivable accounts
 - Employees
 - Printing of employee's weekly paychecks

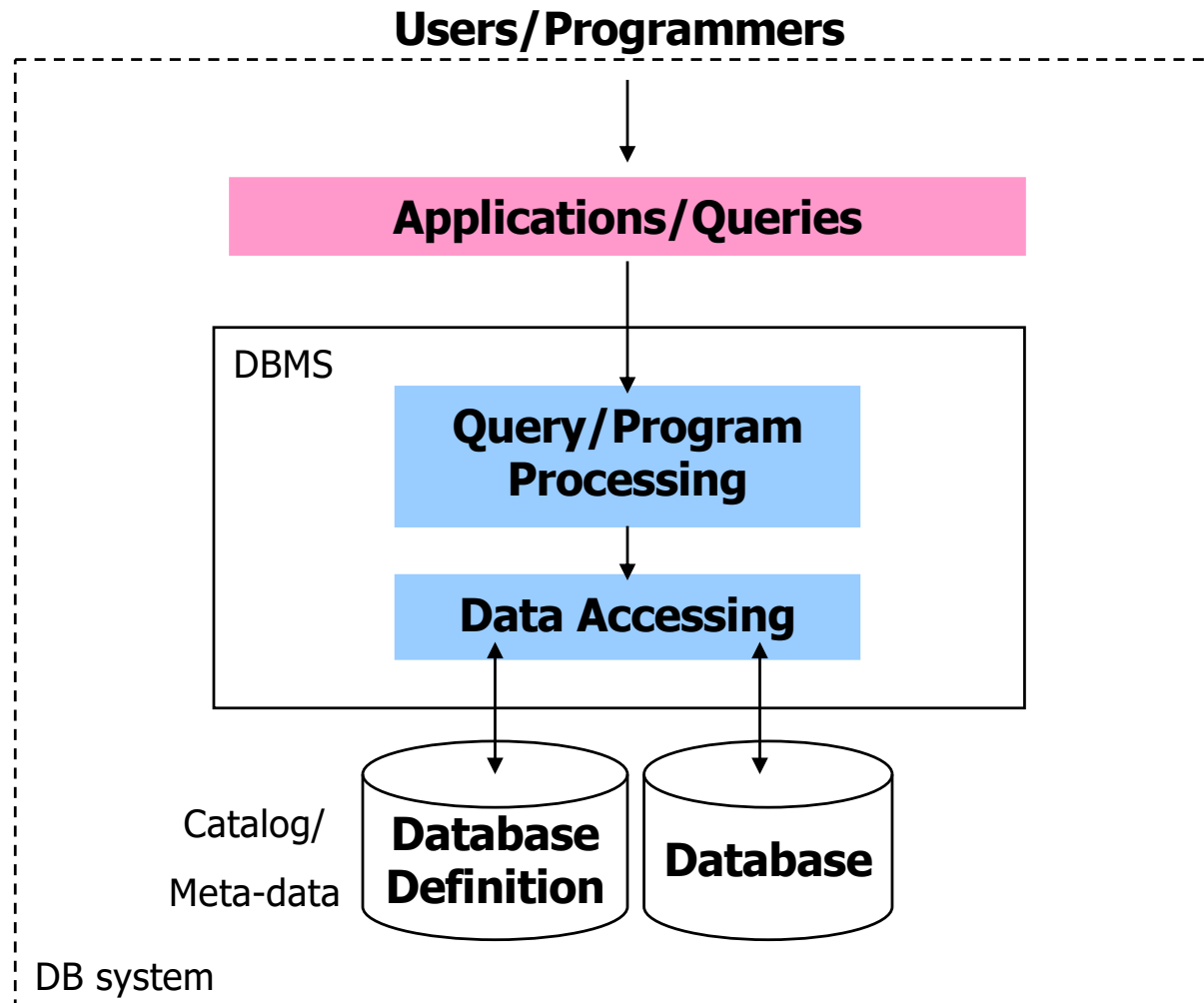
Database (DB)

- Represents some aspect of the real world
- A logically coherent collection of data with some inherent meaning
 - Random assortment of data cannot correctly be a database
- Is designed, built, and populated with data for a specific purpose, for intended group of users or applications

Database Management System

- A collection of programs that enables users to create and maintain a database
- A general-purpose software system that facilitates
 - **Definition** – specifying the data types, structures, and constraints for the data
 - **Construction** – storing the data itself on some storage medium
 - **Manipulation** – querying the database to retrieve data, updating the database to reflect changes, generating reports from the data
 - **Sharing** – allowing multiple users/programs to access the database concurrently

Database system



Source: Elmasri & Navathe, 7E

Example

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

Source: Elmasri & Navathe, 7E

Example

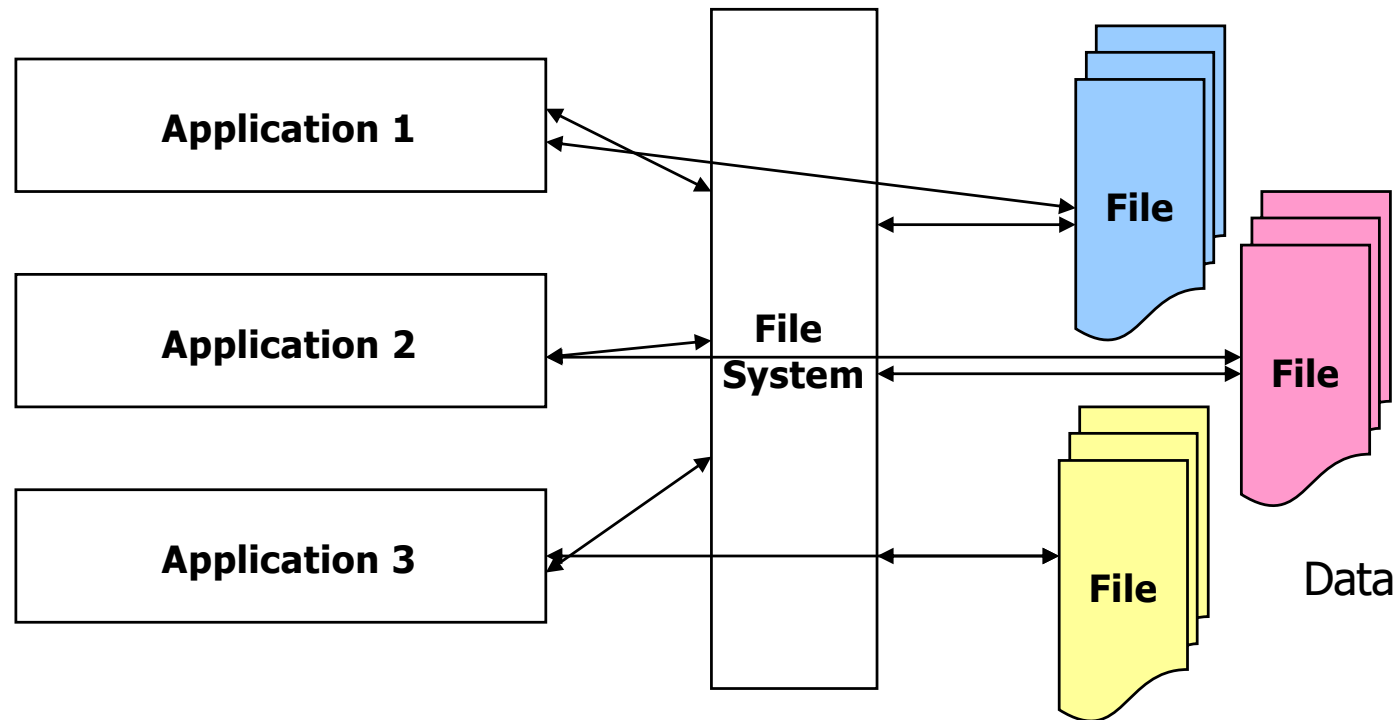
■ University database

- Definition
 - Specify the structure of records (data elements, data types)
- Construction
 - Store data to represent a student, a course, or a section... as a record
- Manipulation
 - Query
 - * Retrieve the transcript of 'Smith'
 - * List the prerequisites of the 'Database' course
 - Update
 - * Change the class of 'Smith' to sophomore
 - * Create a new section for the 'Database' course for this semester

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- **The evolution of database systems**
- Characteristics of the database approach
- Database users
- Architecture of a DBMS
- DBMS category
- Data models
- Database languages

File approach



An application program has its own data

File approach

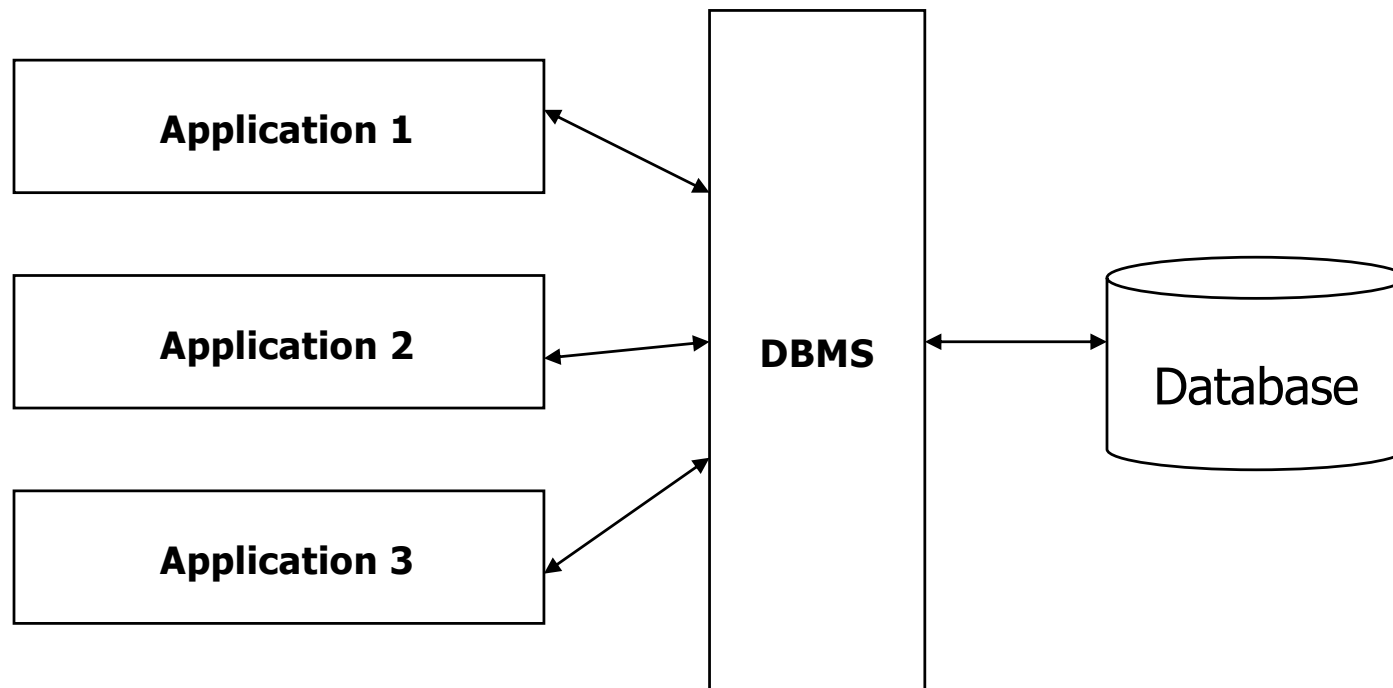
■ Limitations

- Data redundancy
 - Wasted storage space
 - Opportunities of the inconsistency
- Data sharing is limited
- Difficult recovery
- Low security

■ But, still be used in some applications

- Small size DB
 - Storing and accessing data only, not including other processing operations
- Fee costs less
 - Operation or maintenance

Database approach



Content

- Introduction
- The evolution of database systems
- **Characteristics of the database approach**
 - Self-describing
 - Insulation between programs and data
 - Data abstraction
 - Views of data, sharing of data
- Database users
- Architecture of a DBMS
- Properties of DBMS
- Data models
- Database languages

Self-Describing

- The DB system contains not only the DB itself, but also a complete definition/description of the DB structure
- The definitions are stored in ***catalog***
 - Contains information such as the structure of data, type and storage format of data items, and constraints on the data
- Information stored in catalog is called ***meta-data*** (data of data)
- Many applications can access to the DB
 - Refer to catalog, knowing the structure of files in specific DB (type and format of data)

Self-Describing

RELATIONS

Relation_name	No_of_columns
STUDENT	4
COURSE	4
SECTION	5
GRADE_REPORT	3
PREREQUISITE	2

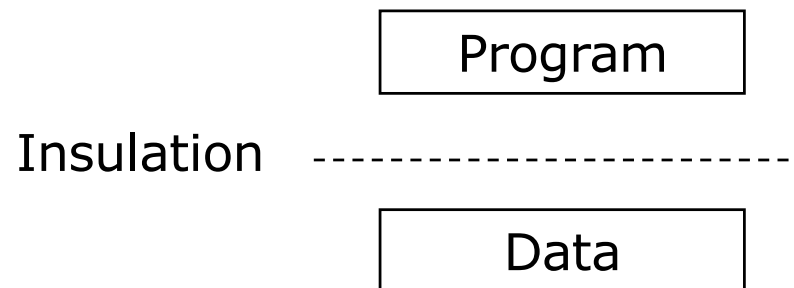
COLUMNS

Column_name	Data_type	Belongs_to_relation
Name	Character (30)	STUDENT
Student_number	Character (4)	STUDENT
Class	Integer (1)	STUDENT
Major	Major_type	STUDENT
Course_name	Character (10)	COURSE
Course_number	XXXXNNNN	COURSE
....		
....		
....		
Prerequisite_number	XXXXNNNN	PREREQUISITE

Source: Elmasri & Navathe, 7E

Insulation

- The structure of data is stored in *catalog* separately from the access programs
 - Program-Data independence



- A little change in the structure happens
 - Application programs are rarely revised

Data abstraction

- The DB system provides a ***conceptual representation*** of the data to hide certain details of
 - How the data are stored and maintained
 - How the operations are implemented

- ***Data model*** is a type of data abstraction
 - Logical concepts
 - Objects
 - Properties
 - Relationships

→ Easier for user to understand than computer storage concepts

Views of data

- A database has many users
- Each user may require a different ***perspective or view*** of the database
- A view may be
 - A subset of the database
 - Aggregate data that are derived from the database

Sharing of data

- A multiuser DBMS
 - Allows users to access the DB at the same time
 - Data for many applications are integrated and maintained in a *single DB*
- Using concurrency control mechanisms to access the data reasonably
 - Avoid data contention
 - Ensure the data will always be valid when they are accessed

Content

- Introduction
- The evolution of database systems
- Characteristics of the database approach
- **Database users**
 - Database administrator (DBA)
 - Database designer
 - End user
- Architecture of a DBMS
- DBMS category
- Data models
- Database languages

Database administrator

- Many people use the same resources
 - Need a chief administrator to oversee and manage
- Responsibility
 - Administering the DB
 - Authorizing access to the DB
 - Coordinating and monitoring the use of DB
 - Acquiring software and hardware resources as needed

Database designer

- Responsibility

- Identifying the data to be stored in the DB
- Choosing appropriate structures to represent and store the DB
- Communicating with all DB users to understand their requirements, to come up with a design that meet the requirements

- Can be

- Staff of the DBA
- Other staffs taking responsibilities after the DB designed is completed

End user

- People whose jobs require to access to the DB
 - Querying, updating, generating reports
- Categories
 - Casual end user
 - Naïve or parametric end user
 - Sophisticated end user

End user

- People whose jobs require to access to the DB
 - Querying, updating, generating reports

- Categories
 - Casual end user
 - Occasionally access the DB
 - Need different information each time
 - Use sophisticated DB query language to specify requests
 - Middle or high-level manager
 - Naïve or parametric end user
 - Sophisticated end user

End user

- People whose jobs require to access to the DB
 - Querying, updating, generating reports

- Categories
 - Casual end user
 - Naïve or parametric end user
 - Constantly query and update the DB
 - Use standard types of queries and updates that have been programmed and tested
 - Employee
 - Sophisticated end user

End user

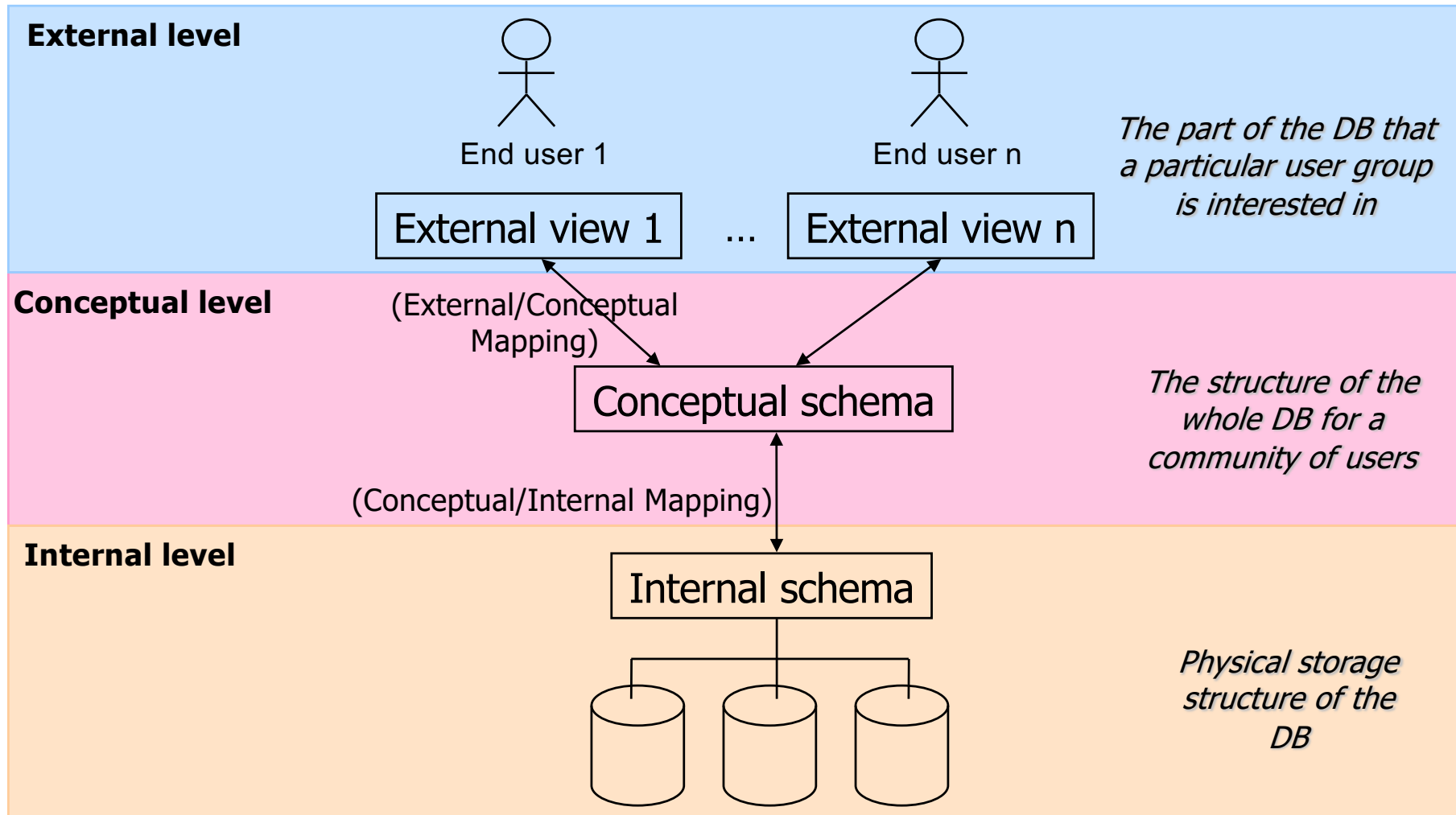
- People whose jobs require to access to the DB
 - Querying, updating, generating reports

- Categories
 - Casual end user
 - Naïve or parametric end user
 - Sophisticated end user
 - Be familiar with the facilities of the DBMS
 - Implement the applications to meet the complex requirements
 - Engineers, scientists, business analysts

Content

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- **Architecture of a DBMS**
- Properties of DBMS
- DBMS category
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- Database languages

Three-schema architecture



Source: Elmasri & Navathe, 7E

Data independence

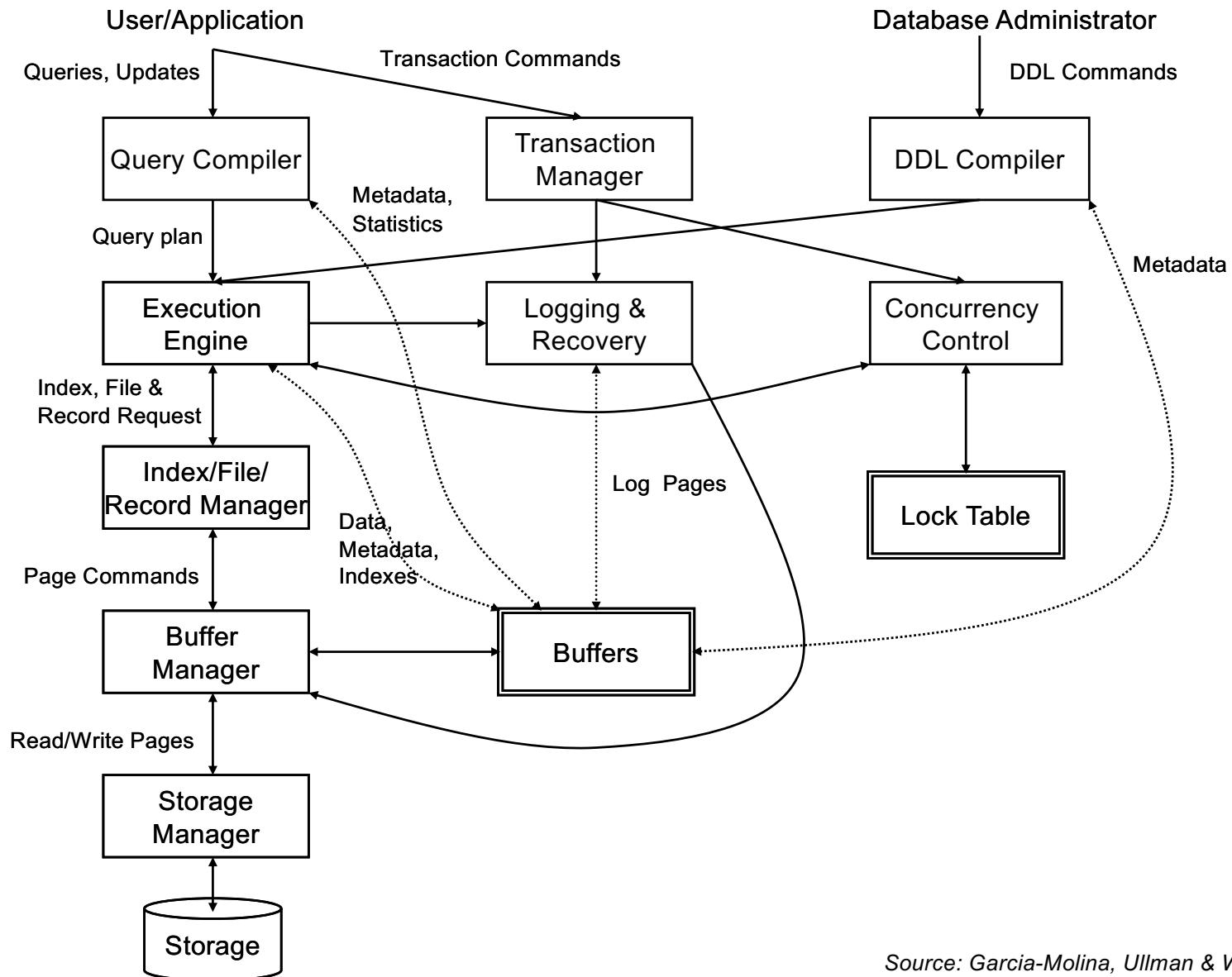
- **Logical data independence**

Can we add a new column or attribute without rewriting the application?

- **Physical data independence**

Do you need to care which disks/machines are the data stored on?

Components



Source: Garcia-Molina, Ullman & Widom, 2E

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Relational DB is not enough

- Newer applications require characteristics
 - More complex structures for stored objects
 - New data types for storing
 - Graphic images, video clips, sound waveforms
 - Large document, large object (LOB)
 - Longer duration transactions
 - Nonstandard operations
- Lead to the development of additional *semantic data modeling* concepts

Categories

- Relational database
- Object-oriented database
- XML database

- NoSQL database
 - “Not only SQL” : not use the tabular relations as relational databases
 - Non-relational, distributed, open-source
 - Horizontally scalable

- Database vs Data lake

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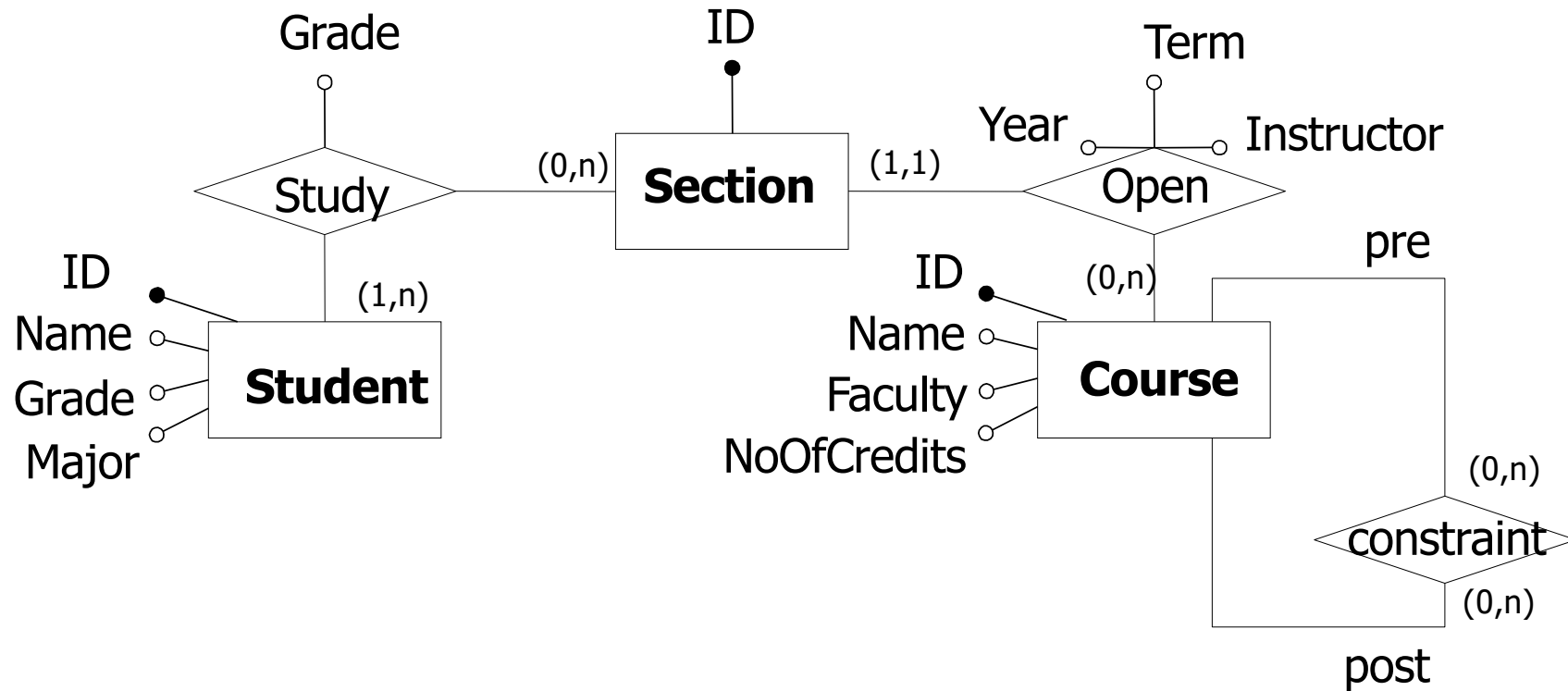
Definition

- A collection of ***concepts*** that can be used to describe the ***structure of a DB***
 - Data types, relationships, and constraints
- Including a set of basic ***operations*** for specifying retrievals and updates on the DB

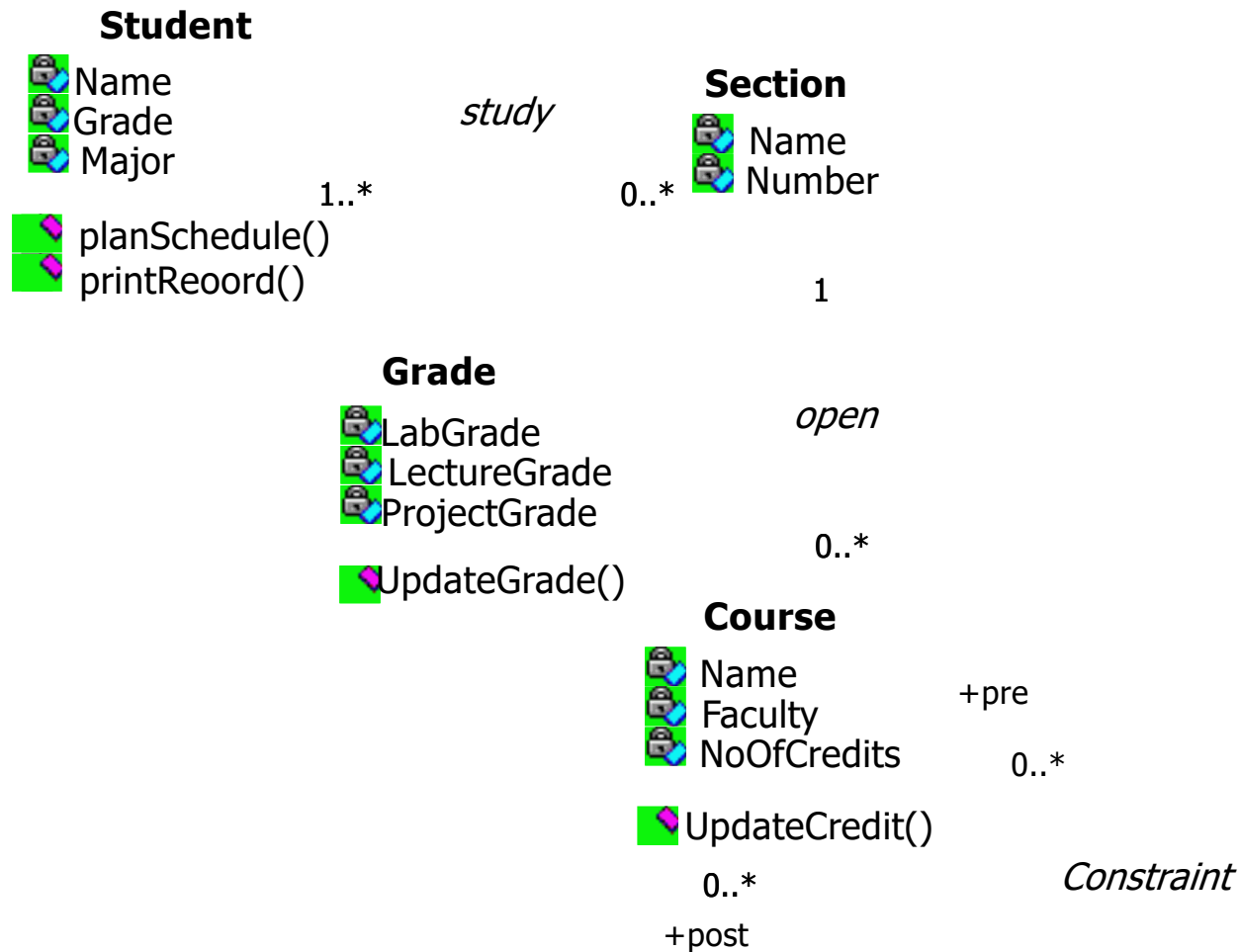
Categories

- High level data model
 - Provide concepts that are close to the way users perceive data
 - Eg: entity relationship model, object-oriented model...
- Implementation data model
 - Provide concepts that may be understood by end users, but that are not too far from the way data is organized within the computer
 - Eg: relational model, network and hierarchical models...
- Low level data model
 - Provide concepts that describe the details of how data is stored in the computer

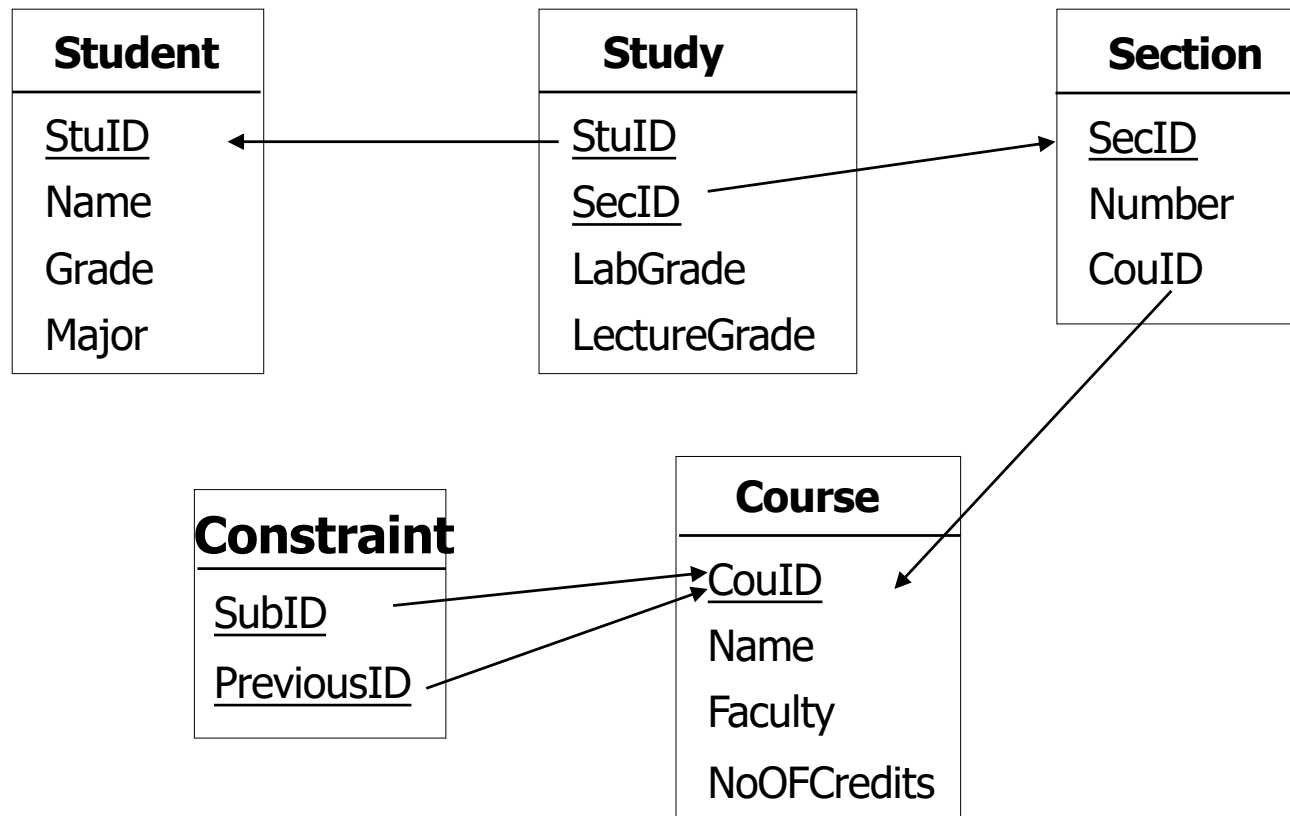
Example of ER model



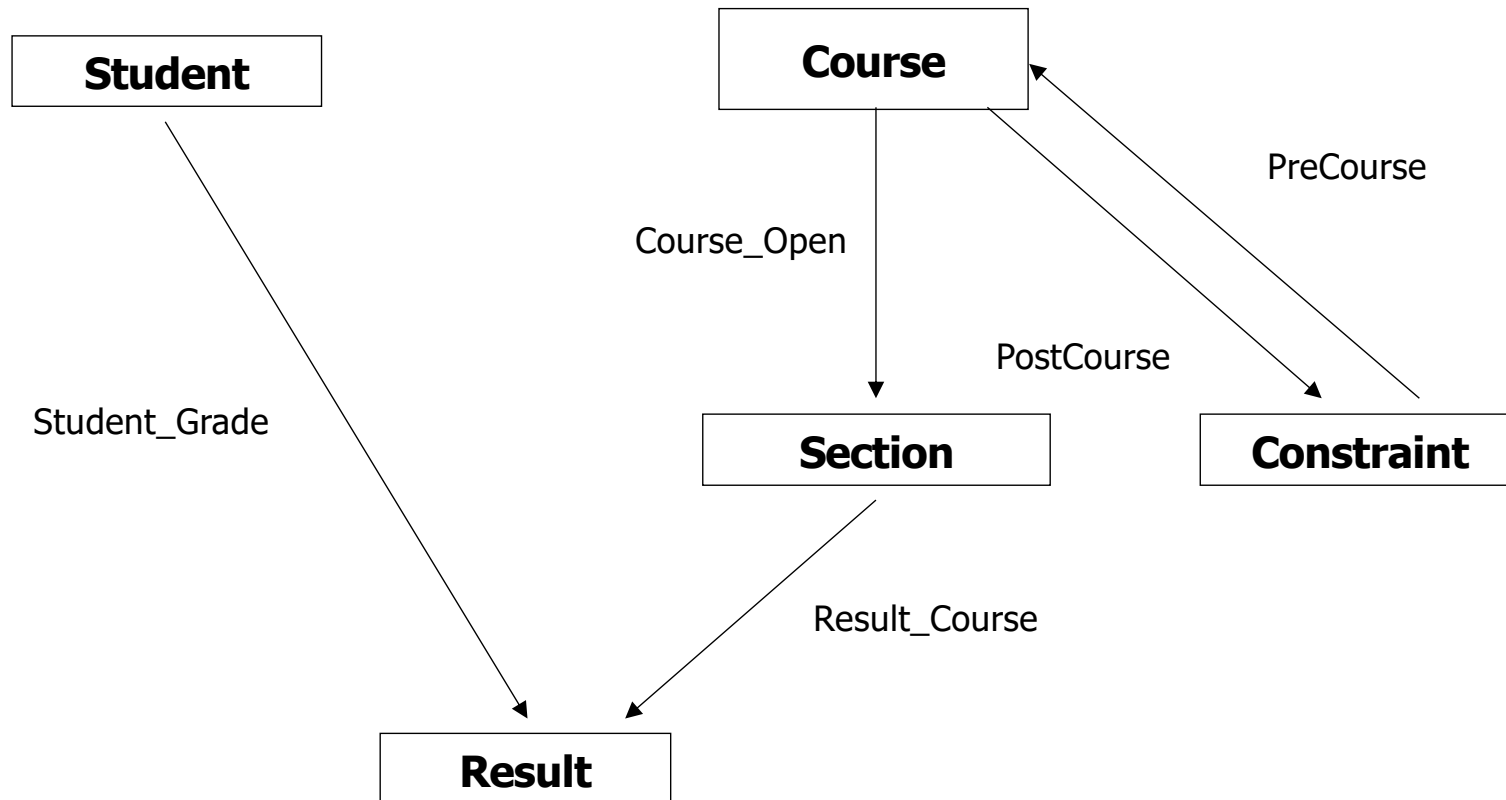
Example of Object-Oriented model



Example of relational model



Example of network data model



Example of hierarchical data model

Level 1:

Result	
LabGrade	LectureGrade

Level 2:

Section	
Name	Number

Student		
Name	Grade	Major

Level 3:

Course		
Name	Faculty	NoOfCredits

Content

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- **Database languages**

Relational database languages

- DDL – Data Definition Language
 - Identify descriptions of the schema constructs
 - Store the schema description in the DBMS catalog
- SDL – Storage Definition Language
 - Specify the internal schema and the mappings between two schemas
- VDL – View Definition Language
 - Specify user views and their mapping to the conceptual schema

Relational database languages

- DML – Data Manipulation Language
 - Provide a set of operations including retrieval, insertion, deletion and modification of the data
 - Two types
 - High level (nonprocedural)
 - * Entered interactively from a display monitor/terminal
 - * Embedded in a general-purpose programming language
 - Low level (procedural)
 - * Must be embedded in a general-purpose programming language

OODB languages

- ODL – Object Definition Language
 - A standardized language for specifying the structure of database in object-oriented terms
- OQL (Object Query Language)
 - A standardized query language for object-oriented database
 - Combines SQL with object-oriented programming paradigm to operate on data described in ODL

XML database languages

- DTD – Document Type Definition
 - A set of rules defining the structure and the legal elements and attributes of an XML document
- XML schema
 - Defining elements and attributes that appear in the XML document
- XPath and XQuery
 - Navigating parts in XML documents
 - Finding and extracting elements and attributes

NoSQL database languages

- UnQL (Unstructured Query Language)
 - Couchbase, SQLite
- CQL (Cassandra Query Language)
- ...

Discussion

- When will we use the DB approach?
- When will we not use the DB approach?

Next lecture

- ER model
- Read Chapter 3 [Fundamentals of Database System, 7E], from the page 59th
- Watch a video clip

